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Fusing deep learning features of triplet leaf image patterns to boost soybean cultivar identification

Bin Wang[a,b,*], Hao Li[a], Jiawei You[a], Xin Chen[b], Xiaohui Yuan[c], Xianzhong Feng[d]

[a] School of Information Engineering, Nanjing University of Finance & Economics, Nanjing 210023 China
[b] Institute for Integrated and Intelligent Systems, Griffith University, QLD 4111, Australia
[c] School of Plant Protection, Jilin Agriculture University, Changchun 130118, China
[d] Northeast Institute of Geography and Agroecology, Chinese Academy of Sciences, Changchun 130102, China

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ABSTRACT

Soybean cultivar recognition plays a vital role in cultivar evaluation, selection and production. Recently, there is an increasing interest in taking leaf image patterns as clues for distinguishing soybean cultivars. However, due to the higher inter-class similarity of soybean cultivars over plant species, the cultivar classification accuracies reported by the existing methods are far lower than those published on plant species recognition which make computer vision community have a concern whether leaf image patterns can provide sufficient discriminative information for identifying soybean cultivars. In this paper, we explore fusing deep learning features of leaves from different parts of soybean plants for achieving an accurate cultivar recognition. In our method, the deep learning features of triplet leave image patterns that consists of leaves from the lower, middle, and upper parts of soybean plants are fused by two methods, distance fusion and classifier fusion. In the former, the L_1 distance measurements defined on the deep feature spaces of triplet leaf image patterns are fused prior to using 1NN classifier for classification. While in the later, the SVM classifiers trained by the deep features of triple leaf image patterns are combined by sum rule for cultivar prediction. We use the SoyCultivar200 leaf dataset which consists of 6000 samples from 200 soybean cultivars as benchmark. Our method achieves an exciting classification rate of 83.55% which demonstrates that our proposed fusion of deep features of triplet leaf image patterns can provide strong discriminative information for accurately identifying soybean cultivars.

1. Introduction

As a species of legume herb, soybean or soya bean is a globally important crop, providing oil and protein, and is treated as the main crop in many countries. For instance, Soybean crops are the main agricultural product for export and have great importance in the Brazilian economy (Moreira et al., 2015). While in the United States, soybeans are an extremely important crop providing the largest source of animal protein feed and the second largest source of vegetable oil (Liang et al., 2018). Due to the growing population under current adverse environmental conditions, the research on soybean breeding, growth, development, and yield for providing enough food is of particular significance.

Soybean has many cultivars (varieties) and experts in agriculture also continue to develop new cultivars with high productivity and profitability to the soybean industry. Under low-input and organic farming conditions, cultivar choice and breeding are regarded as major tools for improving crop performance (Lammerts van Bueren and Myers, 2012). The identification of soybean cultivars accordingly plays a vital role in the evaluation, selection, and production of soybean varieties (Cavassim et al., 2013). In additional, soybean cultivar recognition also contributes to the study of plant phenotype (Minervini et al., 2016).

Leaves are one of the important organs for botanists to distinguish plant species. In computer vision community, many researchers (Hearn, 2009, Kumar et al., 2012, Wang et al., 2015) have made great progress on using leaf image patterns for developing automatic plant species recognition systems. There are many methods utilizing leaf features such as shape, texture, vein features, or their combination for species-level recognition and achieved high classification accuracy rates (more than 90% for some algorithms) on some published leaf image datasets like

* Corresponding author at: School of Information Engineering, Nanjing University of Finance & Economics, Nanjing 210023 China.
E-mail addresses: wangbin@nufe.edu.cn (B. Wang), 804421401@qq.com (H. Li), 973700530@qq.com (J. You), xin.chen12@griffithuni.edu.au (X. Chen), yuanxh@gooalgene.com (X. Yuan), fengxianzhong@iga.ac.cn (X. Feng).

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Leafsnap (Kumar et al., 2012), Leaf100 (Wang et al., 2014), ICL (Hu et al., 2012), MEW2012 (Novotný and Suk, 2013), etc. Then we have a concern about whether leaf image patterns can be used as main clues for distinguishing soybean cultivars. This is an interesting yet challenging problem. In species level, many plants differ from their leaf patterns (see the first row in Fig. 1). However, in cultivar level, since different soybean cultivars belong to the same species, their high inter-class similarities (see the second row in Fig. 1) make it very tough even for experts to distinguish them by their leaf patterns.

Recently, there are an increasing number of efforts (Larese and Granitto, 2016; Baldi et al., 2017; Wang et al., 2020; Chen and Wang, 2020) on using leaf image patterns as clues for cultivar-level recognition. Their studies on soybean cultivar classification confirm the availability of cultivar information in leaf patterns in some extent. However, due to the higher inter-class similarity of soybean cultivars over plant species, the classification accuracies reported by the existing cultivar recognition methods are far lower than those published on plant species recognition which make computer vision community have a concern whether leaf image patterns can provide sufficient discriminative information for identifying soybean cultivars.

In this paper, we focus on soybean cultivar recognition which is known as a very fine-grained pattern recognition problem. Different from the existing soybean cultivar recognition methods that take traditional hand-crafted features into account, our proposed method pay attention on using deep learning techniques for automatically capturing discriminative patterns from soybean leaf images. The main contributions of this work are: We propose to use two feature fusion methods, named distance fusion and classifier fusion, respectively, for effectively integrating deep features of triplet leave image patterns that consists of leaves from the lower, middle, and upper parts of soybean plants. We conduct experiments on the challenging SoyCultivar200 leaf dataset (Wang et al., 2020) and achieves an exciting classification accuracy of 83.55% which greatly improves the previously reported classification score by 11.15%. The encouraging experimental results indicates the potential of our proposed deep features fusion of triplet leaf image patterns for cultivar recognition.

The rest of this paper is organized as follows: In Section 2, the typical algorithms that use leaf image patterns for identifying plant species and cultivars are reviewed respectively. Section 3 presents the details of our proposed deep features fusion method for soybean recognition. Our experimental study is presented in Section 4, the results of our proposed method compared with the state-of-art benchmarks on the SoyCultivar200 leaf dataset are reported. Conclusions are drawn in Section 5.

2. Related work

According to the identifying granularity, the existing leaf recognition methods can be categorized into species-level and cultivar-level. Some representative methods of them are surveyed in this section.

2.1. Species-level recognition

In the past decades, many methods have been proposed to charactering leaf image patterns for distinguishing plant species. They can be classified into traditional handcrafted methods and deep learning methods. In traditional handcrafted methods, the leaf visual characteristics, such as leaf shape, texture, vein and colour, are manually extracted as cues for identifying plant species. While the later directly learn discriminative features from the raw representation of leaf images using the training samples of leaf images.

Handcrafted features. Many plant species differ from their leaf shapes which provide significant clues for botanists to identify species (Cope, Corney, Clark, Remagnino, & Wilkin, 2012). A large body of leaf shape descriptors have been proposed and achieve encouraging recognition accuracies. Some of them focus on depicting the boundary characters of leaf images. Hierarchical string cuts (HSC) (Wang and Gao, 2014) characterizes a contour segment using the spatial distribution information of its all points relative to its string that cuts the segment. It has low time complexity and is suitable for large shape database retrieval with competitive accuracy. Various curvature measurements such as Integral Invariants (Kumar et al., 2012), Multiscale Arch Height (Wang et al., 2015), Triangular Representation (Mouine et al., 2013) and so on have been designed for depicting the boundary property of leaves. Besides the boundary descriptors, there are also some methods capturing the interior region features of leaf shapes, like Image Moments (Novotný and Suk, 2013; Wang et al., 2008), 2D Fourier Power Spectrums (Horaisová and Kukal, 2016), and Chord Bunch Walks (Wang et al., 2019) which also achieve desirable results on plant species recognition.

Leaf texture is another important clue for distinguishing plant species. The popular texture descriptors such as Gabor Filter (Vijaya-Lakshmi and Mohan, 2016; Cope et al., 2010), Gray-level Co-occurrence Matrix (Tang et al., 2015), Local Binary Pattern (LBP) (Naresh and Nagendraswamy, 2016), and Pairwise Rotation Invariant Co-occurrence LBP (Qi et al., 2014) have been applied for leaf-based plant species recognition.

As a special structure pattern, leaf vein can be used as a complimentary clue for further improving the accuracy of species classification. Charters et al. (2014) designed a new descriptor named Eagle which characterises the overall venation structure using the edge patterns among neighbouring regions. Their experimental results show that the EAGLE descriptor can boost the performance of the SURF local descriptor by 6% on the Swedish leaf image dataset. Larese et al. (2014b) proposed to compute a lot of morphological features on the segmented venation and shown its effectiveness on legumes species classification.

Deep learning features. With the rapid progress of deep learning techniques, many efforts have been made to automatically extracting leaf features in end-to-end learning for plant species recognition (Hu et al., 2018). Compared with the traditional feature extraction modules, the deep learning techniques are fully data-driven which make them

Fig. 1. Species versus cultivar of plants: An example to show the high inter-class similarity of cultivars (the second row) compared with the large inter-class variety of species (the first row). The first row: Example leaves from different species of the Leaf100 dataset (Wang et al., 2014). The second row: Example leaves from different cultivars of the SoyCultivar200 dataset (Wang et al., 2020).

generally have higher possibilities to capture discriminative feature patterns when there is enough training data. Grinblat et al. (2016) used a deep convolutional neural network (CNN) to extract leaf vein features for classifying three different legume species: white bean, red bean, and soybean. A mean accuracy of 96.90% attained by the 5-layer model was reported in their experimental study.

Lee et al. (2017) studied the use of deep learning techniques for plant identification. Their experimental results show that learning the features using CNNs can better describe leaf image patterns than using hand-crafted features. Their study gains two intuitions: (1) venation is a more powerful discriminative feature than shape, and (2) deep learning can help discover more efficient discriminating features than those used for plant identification in previous studies. Shah et al. (2017) proposed a dual-path CNN to jointly learn complementary shape and texture features in each path and optimize them for leaf classification. Their experimental results on three large public datasets demonstrates that the dual-path CNN outperforms the state-of-the-art including CNN-based methods and handcrafted features.

To provide plant leaf recognition at multiple scales, Hu et al. (2018) propose a multiscale fusion CNN (MSF-CNN) in which an input image is down-sampled into multiple low resolution images that are further fed into the MSF-CNN architecture to learn features at different depths. The features of two different scales are fused by a concatenation operation. The last layer of MSF-CNN integrates all discriminative information for plant species prediction. Tan et al. (2020) proposed a new CNN-based method, named D-leaf for identify plant species. They use three different CNN models, pre-trained AlexNet, fine-tuned AlexNet and D-leaf, for leaf vein feature extraction and five machine learning techniques, SVM, ANN, k-NN, NB and CNN for classification. Their experimental results show that D-leaf can be an effective automated system for plant species identification. Bisen (2021) proposed an automated plant identification system based on CNN and attained an accuracy of 97% on classifying 15 tree species of the public Swedish leaf dataset.

2.2. Cultivar-level recognition

Using leaf image patterns as clues for identifying plant species has achieved great successes in the past decades. Recently, there is an increasing concern whether leaf image patterns can also provide powerful discriminative information for cultivar-level recognition. Larese et al. (2014a) proposed to characterize and analyze the leaf venation network for distinguishing legume varieties. They reported an average accuracy of 60.20% on distinguishing three diverse soybean cultivars using the best version of their proposed methods. Oleander (Nerium oleander L.) has many cultivars and it is very difficult to distinguish them. Baldi et al. (2017) developed a Backpropagation Neural Network (BPNN) based on analysing oleander image patterns as support tool for identifying oleander cultivars, where the BPNN model is built using 18 morphometric and colorimetric leaf parameters of 880 leaves from 22 cultivars. Their experimental results reported that the percentage of leaves attributed to the correct class reached 54.55% on total of the analysed leaves of 22 cultivars.

Larese and Granitto (2016) used the combination of Self-Invariant Feature Transform (SIFT), Bag of Words (BoW), and Support Vector Machines (SVM) to detect relevant vein patterns for distinguishing three legumes, red bean, white bean, and soybean, and three cultivars of soybean. According to the experimental results reported by them, the best average accuracy of classifying three soybean cultivars achieves 63.82% when using the dictionary with 100 words. Tavakoli et al. (2021) proposed to use convolutional neural networks (CNNs) for classifying 12 different cultivars of common beans that belong to three various species. In their methods, instead of the standard softmax loss function, the advance loss functions, like Additive Angular Margin Loss and Large Margin Cosine Loss, are adopted for improving the discriminative power. They reported the maximum mean accuracies of 91.37% and 86.87% on the classifications of cultivars from same species and cultivars from different species, respectively.

The existing methods proposed for leaf recognition only consider single leaf pattern and ignore the complementary properties of leaves from different parts of plants. Wang et al. (2020) made the first attempt of using joint leaf patterns from the lower, middle and upper parts of soybean plants for soybean cultivar identification. In their study, a novel multiscale sliding chord matching (MSCM) approach is developed for characterizing single leaf pattern and the descriptors of the leaves of the lower, middle and upper parts are jointly used for enhancing the discriminative power of leaf image descriptors. They built a cultivar leaf dataset, SoyCultivar200, consisting of 6000 samples from 200 soybean cultivars for classification performance tests and when using the single leaf image pattern for cultivar classification, their method achieved the identification accuracies of 34.70%, 33.50, and 31.00% for the leaf images of lower, middle and upper parts, respectively and when using joint leaf image patterns, their method achieved the accuracy of 72.40%.

3. The proposed method

In this section, the details of our proposed deep features fusion methods for soybean cultivar recognition are presented. We begin with building leaf deep feature extractors. Two features fusion methods, distance fusion and classifier fusion, are then proposed for integrating the deep features of triplet leaf image patterns for distinguishing soybean cultivars.

3.1. Building leaf deep feature extractors

Convolutional neural network, also known as CNN, has shown great success in numerous visual recognition tasks. Many publicly available pre-trained CNN models have been actively used as generic feature extractors for object recognition and image classification without additional model training procedure (Lee and Lee, 2019, Chen et al, 2021). There are also many approaches focusing on fine-tuning the pre-trained CNN models for specific image classification tasks (Zhou et al, 2017, Cetinic et al, 2018). In this work, both pre-trained CNN models and fine-tuned CNN models are considered as deep feature extractors for investigating the performance of deep feature fusions for soybean cultivar identification. A leaf pre-trained deep feature extractor can be obtained by choosing a pre-trained CNN model and simply removing the last fully-connected layer of the model. We then use the output of the model as deep feature representation.

Fine-tuning a pre-trained CNN to desired performance usually requires enough training data. As a very fine-grained visual categorization problem, soybean cultivar identification task however cannot present sufficient labelled training data due to the expensive annotating cost. While developing a fine-tuning strategy for few-show learning is still an open issue in the research community of deep learning and is not the focus of this paper. Note that this work concentrates on the deep feature fusions of triplet leaf image patterns. Fortunately, many publicly available leaf image datasets such as ICL (Hu et al., 2012), MEW2012 (Novotný and Suk, 2013), Leafsnap (Kumar et al., 2012), etc. that are designed for species recognition have a large body of labelled samples. These datasets obviously have smaller domain difference with the soybean leaf image dataset compared with the ImageNet dataset. Using them to fine-tune a CNN that is pretrained on the ImageNet can accordingly have a potential to reduce the domain difference for making the model better adapt to the soybean cultivar recognition task.

For improving the performance of deep features to soybean cultivar identification, we use a large size of leaf image dataset in which the species of all the leaves have been labelled as a training dataset to fine-tune a pretrained CNN model as shown in Fig. 2. After that, we remove the last fully-connected layer of the model and use the output of the model as deep feature representation. In our experiments, we use ICL leaf image dataset which is a very large size of benchmark and has been popularly used by many studies (Wäldchen and Mäder, 2018) to fine-

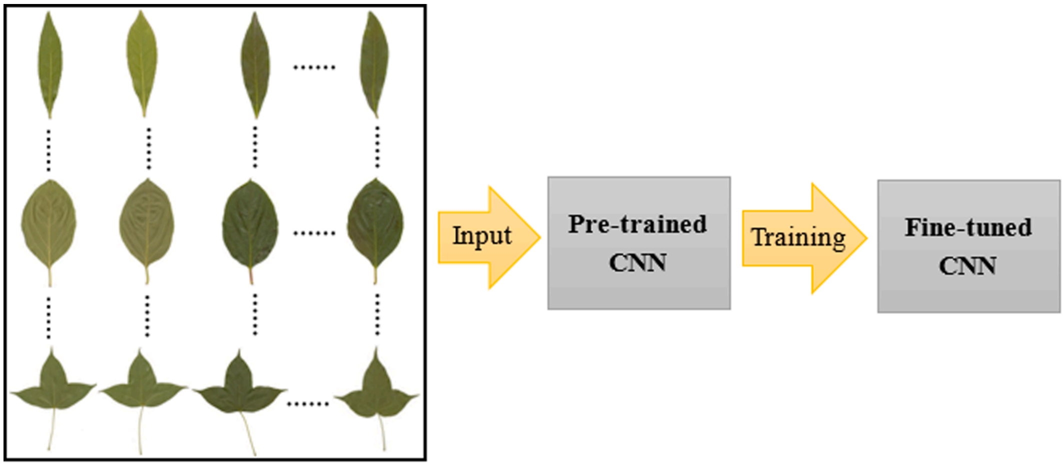

Fig. 2. The pipeline of fine-tuning a pretrained CNN model on a leaf species image dataset that have a large body of labelled samples.

tune the CNN models for building triplet deep feature extractors. Its details are presented in Section 4.3.

3.2. Fusing deep features of triplet leaf image patterns

In this subsection, we propose two feature fusion methods to integrate the deep features of triplet leaf image patterns that consist of leaves from the lower, middle, and upper parts of soybean plants.

Distance fusion. The deep features extracted from a CNN model can be viewed as a feature vector $\mathbf{x}$. Let $x_j^{(i)}$ denote the deep feature vector extracted from the leaf of the i th part of soybean plants of the j th training sample using the CNN model, where $i = 1, 2, 3$ are the indices of the low, middle and upper parts, respectively. Let $\rho(j)$ be a function returning the class label of the j th training sample. Given a testing sample of triplet leaf image pattern, let $y^{(i)}, i = 1, 2, 3$ be its deep feature vectors extracted by the CNN model. The distance between the i th pattern of the test sample and the i th pattern of the j th training sample is defined as

$$d_j^{(i)} = \|y^{(i)} - x_j^{(i)}\|, i = 1, 2, 3, \quad (1)$$

where $\|\bullet\|$ is L_1-norm. The fused distance between the test sample and the j th training sample is defined as

$$D_j = \sum_{i=1}^{3} \frac{d_j^{(i)}}{\overline{d}^{(i)}}, \quad (2)$$

where

$$\overline{d}^{(i)} = \frac{1}{N}\sum_{j=1}^{N} d_j^{(i)} \quad (3)$$

and N is the number of the training samples. Then the class label δ of the test sample is predicted by

$$\delta = \rho\left(\underset{j}{\mathrm{argmin}} D_j\right). \quad (4)$$

The feature fusion proposed above uses the sum of the normalized distance measures of deep features between the test sample and the training sample for feature fusion (see Eq. (2) and Eq. (3)) prior to constructing a 1NN classifier for class label prediction (see Fig. 4). We term it distance Fusion.

Classifier fusion. Besides the proposed distance fusion, we also introduce another feature fusion method, named Classifier Fusion, which is designed to train three different SVM classifiers in advance using triplet leaf image patterns and integrate their classification results for final class label prediction.

Kittler et al. (1998) studied the compound classification where all the pattern representations are used jointly to make a prediction. Their experimental comparisons of various classifier combination schemes indicate that the sum rule outperforms other classifier combination schemes. In the proposed classifier fusion, the widely used SVM is adopted as the basic classifier and the sum rule is used for an combination of different classifiers.

SVM is originally designed for binary classification task. We apply the "one-against-one" approach (Knerr et al., 1990) for multiclass classification. Let M be the number of classes. We can then construct $M(M-1)/2$ classifiers with each training data from two classes. In classification, we use a voting strategy in which each binary classification is treated as a voting to cast a vote to a class and the predicted class is the one that has the maximum number of votes. The more details about this strategy can be obtained from literature (Chang and Lin, 2011). So, a multiclass SVM classifier can be regarded as a combination of multiple binary classifiers.

In the training phase, We first construct triple sets of deep feature vectors, $X^{(i)} = \left\{x_1^{(i)}, x_2^{(i)}, \cdots, x_N^{(i)}\right\}, i = 1, 2, 3$, from the triple leaf image patterns of training set using a deep feature extractor. Then we the triple sets of deep feature vectors to separately train three SVM classifiers, SVM_i, $i = 1, 2, 3$. While in the prediction stage, for a sample to be predicted, we first use the deep feature extractor to yield triple deep feature vectors $y^{(i)}, i = 1, 2, 3$ from its triplet leaf image pattern. Each deep feature vector $y^{(i)}, i = 1, 2, 3$ is then fed into its corresponding classifier SVM_i. Since each class obtains a number of votes, we denote the vote vector that the classifier SVM_i outputs as $P^{(i)} = \left\{p_1^{(i)}, p_2^{(i)}, \cdots, p_M^{(i)}\right\}$, where M is the number of the classes, $p_k^{(i)}$ indicates the number of votes that the k th class receives, and $\sum_{k=1}^{M} p_k^{(i)} = M(M-1)/2$.

Then according to the sum rule, the index of the class label of the test sample is determined by

$$\dot{k} = \underset{k}{\mathrm{argmax}} \sum_{i=1}^{3} p_k^{(i)}, \quad (5)$$

i.e., the index of the class whose total votes from the triplet leaf image pattern is the maximum. Fig. 3 is presented for illustrating the pipeline of predicting the class label of a test sample using the classifier fusion method.

4. Experimental results and discussions

In this section, extensive experiments are conducted to evaluate the

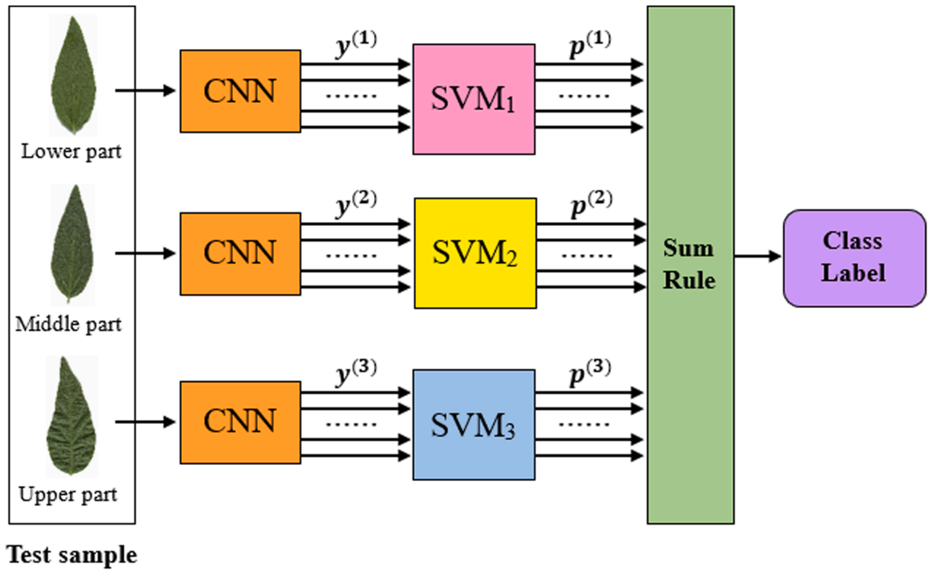

Fig. 3. The pipeline of predicting the class label of a test sample with triplet leaf image patterns using the classifier fusion method.

performance of our proposed method for soybean cultivar recognition. They are divided into two groups. The first is to use single leaf pattern as cues for the identification of soybean cultivars. In this group of experiments, we only use the features of leaves from the single part (lower, middle, or upper parts) of soybean plants for cultivar recognition. The deep feature representations including pretrained deep features and fine-tuned deep features, and the representative handcrafted descriptors are carried out and compared for examining the availability of cultivar information in leaf image patterns and the effectiveness of the proposed fine-tuned strategy for CNN models. The second one is to study the fusions of triplet leaf image patterns for identifying soybean cultivars. In this group of experiments, our proposed methods are conducted to fuse deep features of triplet leaf image patterns for validating their large performance improvements and showing their much more accurate classification of soybean cultivars compared with the state-of-the art approach developed for soybean cultivar recognition.

4.1. Dataset

The leaf dataset, SoyCultivar200[1], used in (Wang et al., 2020) is employed as benchmark dataset in our experiments. It includes 200 soybean cultivars with each having ten samples. Each sample has three leaves collected from the lower, middle and upper parts of different soybean plants which results in a total of 6000 leaves included in the SoyCultivar200 dataset. All the samples grew in Heilongjiang Provence known as a major soybean production area in China. In Figs. 4–6, example leaves from the lower, middle, and upper parts of the soybean plants of 200 cultivars are presented, respectively.

4.2. Evaluation protocol

The same evaluation protocol as used in (Wang et al., 2020) is adopted in our experiments. In the first group of experiments, the SoyCultivar200 dataset is divided into three subsets which contain the leaves from the lower, middle, and upper parts of different soybean plants, respectively. Then each subset has $200 \times 10 = 2000$ leaves that are all from the same part of different soybean plants of 200 cultivars. We conduct cultivar classification on each subset. For each subset, the first five leaves are taken as training samples and the remaining ones are used as testing samples. The classification rate is calculated as the ratio of the number of the correct identification tests to the number of all the tests. Then, we interchange the roles of these two sets, i.e. the first five samples of each cultivar are used as testing samples and the remaining ones are used as training samples to conduct another group of classifications. The final classification accuracy rate is reported as the mean value of the rates of the two groups of classifications.

Different from the first group of experiments in which each sample only has single leaf pattern, the second group of experiments are conducted in which each sample has triplet leaf image patterns consisting of leaves from the lower, middle, and upper of different soybean plants as clues. The proposed deep feature fusion methods as introduced in Section 3.2 are used for soybean cultivar recognition. The same method for measuring the classification accuracy rate as used in the form tests is employed in this group of experiments.

4.3. The selected pre-trained CNN models and their fine-tuning

It is worth noting that this study focuses on developing a method of fusing deep features of triplet leaf image patterns for boosting soybean cultivar recognition and the task of designing a novel CNN model is not the goal of this work. In our experiments, we directly apply the existing pre-trained CNN models to our proposed feature fusion methods. There are many pre-trained CNN models built for image classification and there is no need to try each of them. In this study, two popular and widely used pre-trained models, ResNet-50 (He et al., 2016) and DenseNet-121 (Huang et al., 2017), are utilized in our experiments. The ResNet solves the difficulties in training process of deep CNN, and saturation and degradation of accuracy. While the DenseNet model address the issue of gradient disappearance of deep network, strengthens the propagation of features and encourages feature refuse. They have achieved outstanding performance in many applications (Zhong and Zhao, 2020, Deshpande et al., 2021, Ju et al., 2022, Abbas et al., 2021).

The first attempt is to directly use the pre-trained CNN models as generic feature extractors without fine-tuning. The second attempt, as introduced in Section 3.1, is to fine-tune the pre-trained CNN models on the ICL leaf image dataset (Hu et al., 2012). It contains 16,894 leaf images from 220 species with 26 to 1078 samples per species. We divide the dataset into a training set consisting of 11,954 images and a validation set including 4940 images in a split ratio of 7:3. The parameters used for fine-tuning two pretrained CNN models, ResNet-50 and

[1] Download link: https://github.com/NUFE-AIAG/FGLIR

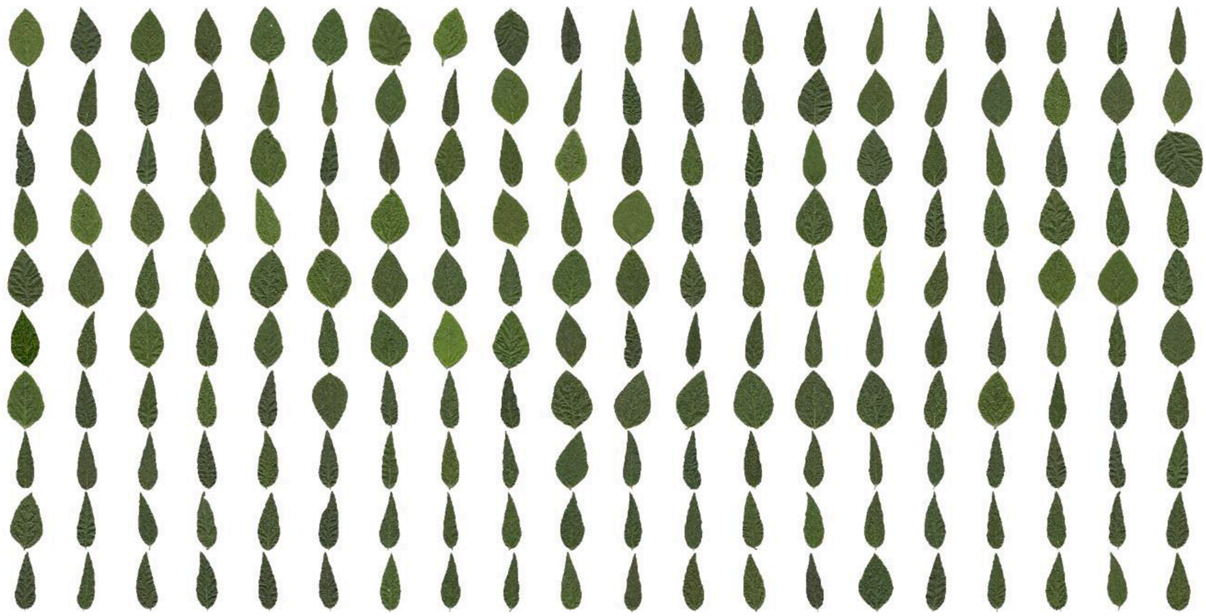

Fig. 4. Example leaves from the lower part of the soybean plants of 200 cultivars (one per cultivar) in the SoyCultivar200 dataset (Wang et al., 2020).

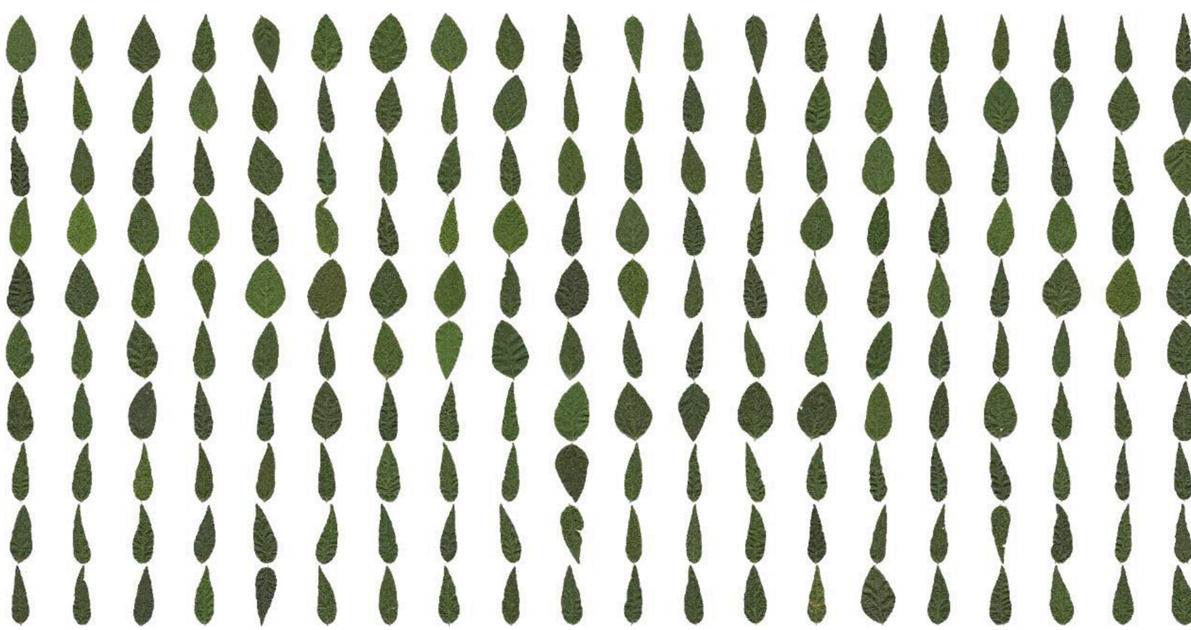

Fig. 5. Example leaves of 200 cultivars (one per cultivar) from the middle part of the soybean plants in the SoyCultivar200 dataset (Wang et al., 2020).

DenseNet-121, are summarized in Table 1. All layers of the pretrained models are released for fine tuning. The lengths of the deep features extracted from ResNet-50 and DenseNet-121 are 2048 and 1024, respectively.

For showing that the obtained fine-tuned ResNet and DenseNet models are not overfitting the data, we plot their loss curves during the training phase in Fig. 7. As can be observed that the distance between the train loss curve and the validation loss curve for each model is very small which shows the overfitting does not occur. We choose the models that achieve the best accuracies on the validation sets as the final fine-tuned models.

4.4. Results and discussion

In the first group of experiments, we evaluate the performances of the deep learning features including pretrained CNN features and fine-tuned CNN features, as introduced in Section 4.3, on using single leaf image patterns for soybean cultivar recognition. 1NN (L_1-norm used) and SVM (linear kernel used) classifications are conducted, respectively, for various deep feature representations. For SVM classification, we use the LIBSVM package (Chang and Lin, 2011) in which the functions, svmtrain and svmpredict, are separately conducted for training and prediction in the proposed algorithm. As for their parameters setting, the parameter t is set to 0 for choosing the linear kernel function in the model training and the remaining parameters are all set to their default values in our experiments, e.g. the parameter e is set to 0 to make the tolerance of termination criterion take the value of 0.001 in the training processing. More details about the functions, svmtrain and svmpredict, and their parameters setting can be found in the literature (Chang and Lin, 2011) and the website "https://www.csie.ntu.edu.tw/~cjlin/

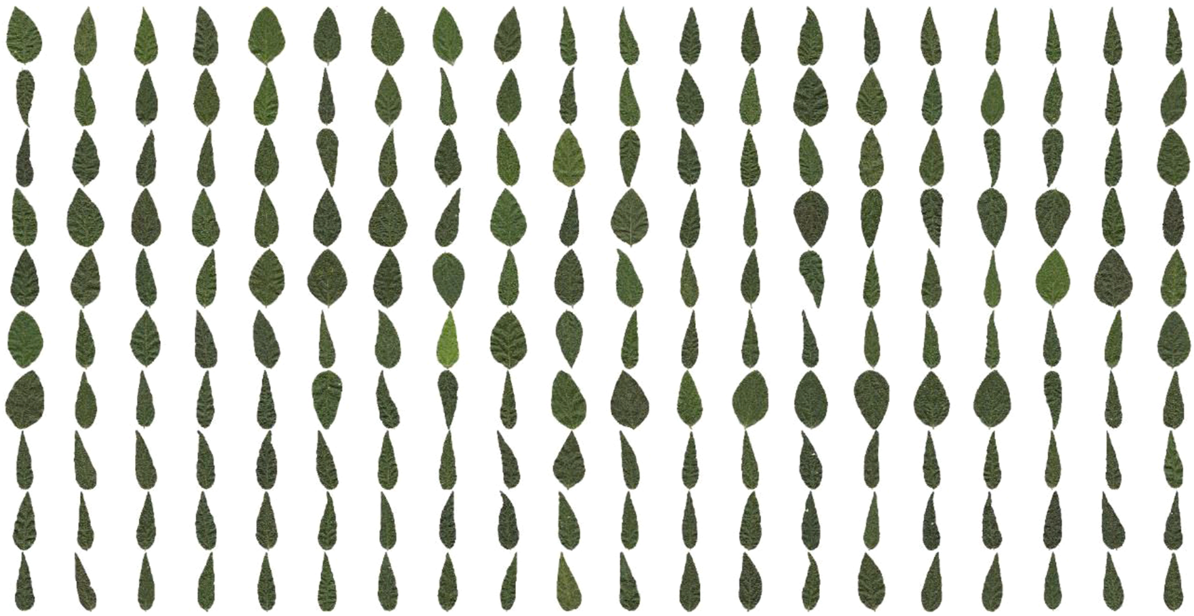

Fig. 6. Example leaves of 200 cultivars (one per cultivar) from the upper part of the soybean plants in the SoyCultivar200 dataset (Wang et al., 2020).

Table 1
The parameters used for fine-tuning the pretrained CNNs, ResNet-50 and DenseNet-121, on the ICL leaf image dataset (Hu et al., 2012) in our experiments.

Parameter	ResNet-50	DenseNet-121
Batch size	32	32
Learning rate	10^{-4}	10^{-4}
Epoch	30	30
Optimizer	Adam	Adam
Loss function	Cross entropy	Cross entropy

libsvm/index.html".

The recent published handcrafted leaf image descriptor, Multiscale Sliding Chord Matching (MSCM) (Wang et al., 2020), which are designed for soybean cultivar identification is also chose as benchmark method. The experimental results of all of the competing methods on the SoyCultivar200 are summarized in Table 2.

It can be seen from Table 2 that the competing handcrafted methods, MSCM (Wang et al., 2020) achieves the score of 34.70% (using leaf image pattern of lower part). While among all the competing deep learning methods, Fine-tuned DenseNet-121 (SVM) achieves the best score of 47.15% (using leaf image pattern of middle part) which outperforms the handcrafted methods by 12.45%. In this group of experiments, we are also very interested in the performance of the proposed fine-tuning strategy on boosting the identification accuracy of soybean cultivars. Through comparing the classification rates of the fine-tuned models and the pretrained models, we can find that all the pretrained models are improved by the proposed fine-tuning strategy on soybean cultivar recognition (Their accuracy increases range from 1.85% to 8.75%). Although restricted by the limited labelled training data, we do not use the soybean leaf images to directly fine-tune the models, the proposed strategy of using a large labelled dataset that has relative small domain difference with the soybean leaf image dataset can also significantly improve the performance on soybean cultivar recognition.

In the second group of experiments, we use the proposed features fusion methods, distance Fusion and classifier fusion, to fuse the deep features of triplet leaf image patterns extracted by the previously used pretrained and fine-tuned CNN models for identifying soybean cultivars. The classification accuracies of them are summarized in Table 3. We also list the classification rate reported by the handcrafted method MSCM (using triplet leaf image patterns) (Wang et al., 2020) as baseline.

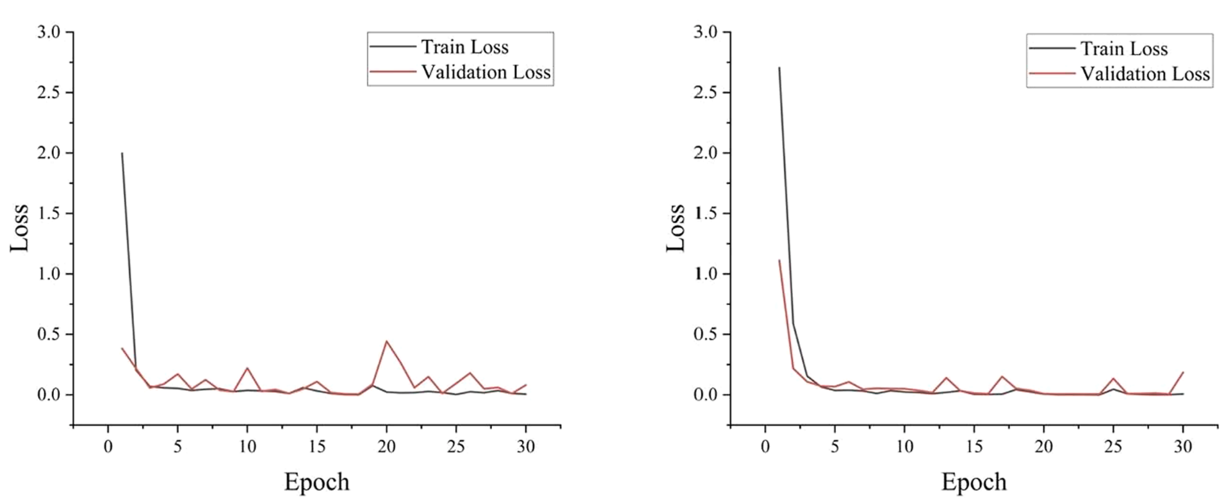

Fig. 7. Loss curves of fine-tuning ResNet-50 (left) and DenseNet-121 (right) on the ICL leaf image dataset (Hu et al., 2012).

Table 2
The classification accuracies (%) of non-feature-fusion methods (using single leaf image patterns) on the SoyCultivar200 dataset.

Algorithm		Lower Part	Middle Part	Upper Part
MSCM (Wang et al., 2020)		34.70	33.55	31.25
Pretrained	ResNet-50 (1NN)	26.60	28.15	21.05
	ResNet-50 (SVM)	35.55	37.30	33.20
	DenseNet-121(1NN)	31.60	35.30	31.80
	DenseNet-121(SVM)	41.60	43.70	39.35
Fine-tuned	ResNet-50 (1NN)	28.80	31.75	29.80
	ResNet-50 (SVM)	37.40	40.10	39.00
	DenseNet-121(1NN)	35.55	38.95	34.40
	DenseNet-121(SVM)	43.50	47.15	44.90

It can be seen that the proposed feature fusion methods achieve an exciting classification accuracy of 83.55% which is 11.15% higher than the score reported by the MSCM method (Wang et al., 2020). Through the comparison of the scores tabulated in Table 2 and Table 3, we can see that using fine-tuned ResNet-50 as feature extractor, the proposed feature fusion makes its accuracy improved from 31.75% to 72.70% for 1NN classification and from 40.10% to 79.00% for SVM classification, respectively. While using fine-tuned DenseNet-121 as feature extractor, the proposed feature fusing method makes its accuracy improved from 38.95% to 76.70% for 1NN classification and from 47.15% to 83.55% for SVM classification. These great improvements of classification accuracies consistently indicate that the proposed fusion of deep features of triplet leaf image patterns has very strong discriminative power to provide much more accurate soybean cultivar identification.

5. Conclusion

In this paper, we have studied the issue of soybean cultivar recognition based on leaf image patterns which is known as a challenging fine-grained pattern recognition problem. Instead of working on handcrafted feature space, we have proposed to fuse the deep learning features of triplet leaf image patterns consisting of leaves from the lower, middle, and upper parts of soybean plants for identifying soybean cultivars. Two methods, distance fusion and classifier fusion, have been designed for deep features fusion of triplet leaf image patterns. We have tested them on the SoyCultivar200 leaf dataset which consists of 6000 samples from 200 soybean cultivars. Our proposed method achieves an exciting classification rate of 83.55% which significantly improves the previously reported classification rate by 11.15% on the SoyCultivar200 leaf dataset. The experimental results demonstrate that the fusion of deep features of triple leaf image patterns can provide strong discriminative information for accurately identifying soybean cultivars which also indicates the potential of the research along this direction.

It is worth pointing out that Soybean cultivar recognition is also a few-shot fine-grained image classification problem. Rather than directly fine-tuning the CNN models on the soybean leaf image dataset, we proposed to fine-tune them on a large leaf image dataset that has relative smaller domain difference with the soybean leaf image dataset than ImageNet dataset. The proposed strategy significantly improves the pretrained CNN models on soybean cultivar recognition. Developing much more effective fine-tuning techniques to address the challenging few-shot soybean cultivar recognition problem is our future work.

CRediT authorship contribution statement

Bin Wang: Conceptualization, Methodology, Formal analysis, Investigation, Writing – original draft, Writing – review & editing, Project administration, Funding acquisition, Supervision. **Hao Li:** Data curation, Validation, Investigation, Visualization. **Jiawei You:** Data curation, Validation, Investigation. **Xin Chen:** Validation, Investigation, Data curation. **Xiaohui Yuan:** Resources, Investigation, Data curation. **Xianzhong Feng:** Resources, Data curation.

Table 3
The classification accuracies of feature fusion methods (using triplet leaf image patterns) on the Soycultivar200 dataset.

Method		Classification Accuracy (%)
MSCM (Wang et al., 2020)		72.40
Proposed Distance Fusion	Pretrained ResNet-50	**65.60**
	Fine-tuned ResNet-50	**72.70**
	Pretrained DenseNet-121	**74.15**
	Fine-tuned DenseNet-121	**76.70**
Proposed Classifier Fusion	Pretrained ResNet-50	**75.90**
	Fine-tuned ResNet-50	**79.00**
	Pretrained DenseNet-121	**80.20**
	Fine-tuned DenseNet-121	**83.55**

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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